Design, Modeling and Safety Engineering of an AI-Assisted Touchless Triage Bot for Emergency Departments

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Abstract

Emergency departments continue to struggle with overcrowding, delayed triage assessments, and variability in acuity scoring. These challenges increase the risk of under-triage and prolonged waiting times for high-risk patients. This project presents a touchless, AI-assisted triage bot designed to support emergency departments by rapidly analyzing vital signs, calculating physiologic indicators, predicting admission risk, applying clinical safety rules, and generating an initial Emergency Severity Index (ESI) level for nurse review. The system collects core vital signs heart rate, respiratory rate, systolic blood pressure, oxygen saturation, temperature, and pain score and derives features such as Shock Index, Hypoxia Flag, Fever Flag, and Tachypnea Flag.

Multiple machine-learning models were developed, including Balanced Logistic Regression, Decision Tree, and XGBoost. SHAP explainability allowed the model to display which vitals influenced risk predictions, improving clinical transparency. A structured QA and Patient Safety Engineering framework guided testing, emphasizing safety-rule overrides (e.g., SBP < 90) and mandatory nurse confirmation. Scenario testing demonstrated that the model identified four out of five sample cases correctly and that the safety rules prevented under-triage by catching borderline critical cases. The results support the feasibility and clinical value of a touchless triage bot that enhances efficiency while ensuring patient safety through human oversight.

Keywords

AI in Emergency Medicine, Clinical Decision Support, ED Triage Automation, Explainable AI, Patient Safety Engineering

Introduction

Emergency departments (EDs) around the world continue to struggle with chronic overcrowding, high patient volumes, and limited staffing resources. These pressures create significant delays in the triage process, which is the first and most crucial step in determining how quickly a patient should receive medical care. When triage assessments are delayed, high-risk patients may wait longer than clinically safe, increasing the likelihood of deterioration, adverse outcomes, and preventable medical complications.

Delayed or inconsistent acuity scoring poses additional clinical risks. Triage nurses often manage many patients simultaneously, and the variability in human judgment can lead to under-triage, where critically ill patients are mistakenly assigned lower priority. Even a short delay in identifying hypotension, hypoxia, severe tachycardia, or other unstable vital signs can have immediate clinical consequences. These challenges indicate a need for additional support systems that can provide fast, consistent, and objective first-pass assessments.

Touchless AI triage offers an emerging solution by allowing patients or automated devices to capture vital signs without requiring manual data entry. This “touchless patient admission” concept reduces workflow bottlenecks and provides immediate physiological analysis before the nurse’s assessment. By computing risk indicators in real time, the system helps clinicians notice abnormalities earlier while maintaining infection control and minimizing contact-based delays.
The objective of this project is to design a real-time, AI-assisted triage bot capable of analyzing vital signs, generating acuity predictions, triggering safety overrides for dangerous patterns, and supporting a nurse-in-the-loop decision-making model. The system automates only the **initial screening** portion of triage. Final ESI assignment always remains under human supervision to ensure clinical accountability. This balance between automation and clinician oversight allows the triage bot to enhance ED efficiency while preserving patient safety.

Literature Review

Tyler, S., Olis, M., Aust, N., Mosier, J., Hill, S., & Thompson, M. (2024). Use of artificial intelligence in triage in hospital emergency departments: A scoping review. *Cureus, 16*(5), e60145. https://doi.org/10.7759/cureus.60145

Tyler et al. (2024) conducted a scoping review examining artificial intelligence applications in emergency department triage across 29 studies, finding that machine learning models consistently outperformed conventional triage systems in predicting patient outcomes. The researchers identified AI-driven systems' capacity to analyze real-time vital signs and presenting symptoms to improve patient prioritization and reduce wait times. However, their analysis revealed persistent implementation challenges including data quality concerns, clinician trust deficits, and algorithmic fairness issues across diverse patient populations. This work establishes that successful AI triage systems require balancing technological capability with practical clinical considerations, validating hybrid AI-human workflows as the most promising approach for emergency medicine settings. The review's emphasis on maintaining human oversight directly informed this project's architecture of combining automated predictions with mandatory clinical confirmation.

Fernandes, M., Vieira, S. M., Leite, F., Palos, C., Finkelstein, S., & Sousa, J. M. C. (2024). Improving triage performance in emergency departments using machine learning and natural language processing: A systematic review. *BMC Emergency Medicine, 24*, 214. https://doi.org/10.1186/s12873-024-01135-2

Fernandes et al. (2024) systematically reviewed 60 studies on machine learning and natural language processing in emergency department triage, revealing that logistic regression served as the most common baseline model due to its interpretability and clinical acceptance, while advanced tree-based algorithms such as XGBoost and LightGBM consistently demonstrated superior performance. The researchers found that ensemble methods captured nonlinear relationships among vital signs and complex interactions signaling early physiological decline more effectively than linear models. Critically, the study noted that incorporating clinical text from nursing notes and patient complaints improved classification accuracy beyond structured vitals alone, though these multimodal approaches introduced additional complexity requiring robust preprocessing pipelines. This tension between performance gains and clinical interpretability

requirements directly shaped this project's decision to compare multiple model architectures, using logistic regression as an interpretable baseline while exploring XGBoost for performance optimization.

Zhao, Y., Wang, Y., Chen, L., & Liu, X. (2022). XGBoost, a novel explainable AI technique, in the prediction of myocardial infarction: A UK Biobank cohort study. *Heart, 108*(23), 1869-1875. https://doi.org/10.1136/heartjnl-2022-321263

Zhao et al. (2022) demonstrated XGBoost's superiority over logistic regression in a large-scale cohort study of over 500,000 patients, achieving an area under the curve (AUC) of 0.86 compared to logistic regression's 0.77 for myocardial infarction prediction. Critically, the researchers addressed interpretability concerns by implementing SHAP values to explain individual predictions, showing that SHAP visualizations revealed which clinical features most influenced each patient's risk score. This work challenges the conventional wisdom that advanced machine learning models necessarily sacrifice interpretability for performance, demonstrating that appropriate explanation frameworks can make complex models transparent for clinical use. The study's integration of high predictive performance with clinical transparency through SHAP directly validates this project's approach of pairing XGBoost with explainability tools to maintain both accuracy and clinician trust.

Jiang, H., Zhang, L., Liu, Y., & Wang, S. (2024). Comparison between XGboost model and logistic regression model for predicting sepsis after extremely severe burns. *Burns & Trauma, 12*, tkae018. https://doi.org/10.1093/burnst/tkae018

Jiang et al. (2024) compared XGBoost and logistic regression for predicting sepsis in severe burn patients, with XGBoost achieving an AUC of 0.91 versus logistic regression's 0.88. The researchers utilized SHAP values to provide clinically interpretable explanations, revealing that XGBoost identified subtle patterns in vital signs and laboratory values indicating early sepsis development that linear models often missed. Importantly, the study emphasized that while XGBoost offered superior predictive performance, the addition of SHAP explanations proved critical for clinical adoption by allowing physicians to understand and verify the reasoning behind high-risk alerts. This work demonstrates that ensemble methods can enhance clinical decision-making in time-sensitive scenarios where early intervention significantly impacts patient

outcomes, supporting this project's hypothesis that combining advanced models with explainability frameworks addresses both performance and trust requirements in emergency medicine contexts.

Lee, J., Kim, H., & Park, S. (2025). Comparison of SHAP and clinician friendly explanations reveals effects on clinical decision behaviour. *npj Digital Medicine, 8*, 58. https://doi.org/10.1038/s41746-025-01958-8

Lee et al. (2025) investigated how explanation methods affect clinician trust and decision-making behavior, comparing SHAP-based explanations with simplified clinical summaries. The researchers found that SHAP visualizations significantly improved clinician confidence in AI recommendations, with physicians demonstrating increased willingness to appropriately act on AI-generated alerts when they could see specific features driving model predictions. This work reveals that transparency in AI systems extends beyond technical requirements to serve as a practical necessity for clinical adoption, as explainability frameworks like SHAP bridge the gap between complex machine learning models and clinical workflows by presenting predictions in formats aligning with clinical reasoning patterns. The study's emphasis on how explanation quality influences clinical behavior rather than merely model accuracy challenges traditional machine learning evaluation paradigms focused solely on predictive metrics, reinforcing this project's commitment to integrating SHAP explanations into the triage interface.

Song, J., Gao, Y., Yin, P., Li, Y., Li, Y., Zhang, J., Su, Q., Fu, X., & Pi, H. (2021). The random forest model has the best accuracy among the four-pressure ulcer adverse events prediction models. *Risk Management and Healthcare Policy, 14*, 1175-1187. https://doi.org/10.2147/RMHP.S297838

Song et al. (2021) compared four machine learning models support vector machines, decision trees, random forests, and artificial neural networks for predicting pressure ulcer adverse events, finding that random forests and decision trees demonstrated superior prediction performance. The researchers' comparative analysis revealed that tree-based models' ability to capture nonlinear relationships and feature interactions made them particularly suitable for healthcare prediction tasks involving complex physiological data. Critically, the study emphasized that model calibration significantly improved performance, with random forests maintaining optimal performance post-

calibration. This work contributes to understanding why tree-based ensemble methods consistently outperform linear models in clinical prediction contexts, supporting this project's inclusion of decision tree as an intermediate comparison point between logistic regression and XGBoost to demonstrate the progression of model complexity and performance gains.

Park, Y., Jackson, G. P., Foreman, M. A., & Gruen, D. (2025). Bias recognition and mitigation strategies in artificial intelligence healthcare applications. *npj Digital Medicine, 8*, 103. https://doi.org/10.1038/s41746-025-01503-7

Park et al. (2025) examined bias recognition and mitigation strategies in healthcare AI, evaluating multiple techniques including SMOTE, cost-sensitive learning, and class weighting for handling imbalanced clinical datasets. The researchers demonstrated that imbalanced datasets cause models to perform poorly on minority classes, which in emergency triage contexts could result in dangerous undertriage of critically ill patients. Their analysis showed that class weighting and cost-sensitive approaches effectively reduced bias without requiring synthetic data generation, while emphasizing that fairness metrics must be monitored continuously across demographic subgroups. This work establishes algorithmic fairness as not merely an ethical consideration but a clinical safety requirement, particularly in high-stakes environments where missed high-risk patients have severe consequences. The study's recommendations directly justified this project's use of balanced class weights and implementation of safety-rule overrides to prevent biased predictions from reaching clinical practice.

Chen, I. Y., Pierson, E., Rose, S., Joshi, S., Ferryman, K., & Ghassemi, M. (2021). Ethical machine learning in healthcare. *Annual Review of Biomedical Data Science, 4*, 123-144. https://doi.org/10.1146/annurev-biodatasci-092820-114757

Chen et al. (2021) provided comprehensive analysis of ethical considerations in healthcare machine learning, warning that models trained on biased historical data produce systematically worse outcomes for underrepresented groups including minorities, older adults, and socioeconomically disadvantaged patients. The researchers argued that technical solutions alone prove insufficient for ethical AI, requiring ongoing collaboration between data scientists, clinicians, and affected communities. They recommended implementing multiple safeguards including diverse training data, fairness-aware algorithms, subgroup performance audits, and

clinical safety rules preventing dangerous predictions regardless of model output. This work synthesizes fairness concerns with practical implementation strategies, establishing that responsible AI development demands equal attention to predictive performance and equitable outcomes across all patient populations. The framework's emphasis on layered safeguards informed this project's multi-pronged approach combining class balancing, continuous fairness monitoring, and mandatory safety overrides.

Vyas, D. A., & Chen, P. (2025). Literature review on the implementation of robotic process automation in healthcare and its social influence. *Procedia Computer Science, 232*, 1545-1554. https://doi.org/10.1016/j.procs.2025.02.208

Vyas and Chen (2025) conducted a systematic literature review on RPA implementation in healthcare, examining both operational benefits and social implications. The researchers documented that while RPA offers significant efficiency improvements and error reduction in administrative tasks, implementation raises ethical concerns including potential job displacement and workforce reskilling requirements. Their analysis emphasized that successful RPA deployment demands balanced approaches considering both technological capabilities and human workforce impacts. Critically, the study argued that automation should free clinical staff to focus on direct patient care rather than replace human judgment in clinical decision-making. This work challenges purely technocentric views of automation, establishing that socially responsible implementation requires explicit consideration of workforce effects and careful delineation between appropriate and inappropriate automation domains. These insights shaped this project's philosophy that automation should enhance rather than replace clinical expertise, reflected in the mandatory nurse confirmation step following automated triage predictions.

Sutton, R. T., Pincock, D., Baumgart, D. C., Sadowski, D. C., Fedorak, R. N., & Kroeker, K. I. (2020). An overview of clinical decision support systems: Benefits, risks, and strategies for success. *npj Digital Medicine, 3*, 17. https://doi.org/10.1038/s41746-020-0221-y

Sutton et al. (2020) examined design principles and implementation strategies for effective clinical decision support systems, identifying that successful CDSS must incorporate transparency, human-centered workflows, and safety overrides preventing incorrect recommendations. The researchers' analysis of CDSS implementations revealed that automated alerts prove effective only when paired

with clinician confirmation and clear escalation pathways to avoid alert fatigue and override behavior. Their work established the hybrid "AI + human review" model as best practice for safety-critical environments like emergency medicine, where automation error consequences can be life-threatening. Importantly, the study emphasized that CDSS effectiveness depends equally on prediction accuracy and seamless integration with existing clinical workflows and electronic health record systems. This comprehensive framework directly informed this project's system architecture combining model predictions with mandatory safety checks and nurse confirmation as essential components for both regulatory compliance and patient safety.

U.S. Food and Drug Administration. (2021). ***Artificial intelligence/machine learning (AI/ML)-based software as a medical device (SaMD) action plan*****. https://www.fda.gov/media/145022/download**

The FDA's (2021) AI/ML-based Software as a Medical Device Action Plan established a comprehensive regulatory framework for AI systems making medical recommendations, emphasizing transparency, predictable model behavior, and human-in-the-loop validation. The action plan outlined five key areas including development of predetermined change control plans for adaptive algorithms, support for good machine learning practices, patient-centered approach incorporating transparency and real-world performance monitoring, and regulatory science methods for algorithm evaluation. Critically, the framework stressed that AI systems in emergency medicine must integrate predictive modeling with explicit risk controls and clinician accountability throughout the total product lifecycle. The guidance's focus on continuous monitoring and change management addresses unique challenges posed by adaptive AI systems that learn from real-world data after deployment. This regulatory context validated this project's system architecture incorporating model predictions, safety overrides, and mandatory nurse confirmation as essential components aligning with FDA expectations for medical device software, while the emphasis on transparency informed the integration of SHAP explainability features.

International Organization for Standardization. (2019). ***ISO 14971:2019 Medical devices Application of risk management to medical devices*****. https://www.iso.org/standard/72704.html**

ISO 14971:2019 established the international standard for risk management processes in medical devices, specifying systematic frameworks for identifying hazards, estimating and evaluating

associated risks, implementing controls, and monitoring effectiveness throughout the entire device lifecycle. The 2019 revision introduced significant enhancements including expanded focus on benefit-risk analysis, strengthened requirements for production and post-production activities, and closer alignment with regulatory requirements in major markets worldwide. The standard's risk management process applies from initial conception through ultimate decommissioning, covering all risk types including data security, usability, and system reliability concerns particularly relevant to software as a medical device. ISO 14971's emphasis on systematic hazard identification and risk mitigation directly informed this project's safety engineering approach, including the implementation of rule-based safety overrides that prevent dangerous predictions from reaching clinical practice regardless of model outputs. The standard's requirement for continuous monitoring and post-production surveillance aligned with the project's design incorporating audit logging and performance tracking mechanisms essential for maintaining patient safety in deployed clinical systems.

Graber, M. A., & Lee, T. (2022). Machine learning in emergency medicine: Promise, barriers, and the path to clinical adoption. *Academic Emergency Medicine, 29*(7), 845-856. https://doi.org/10.1111/acem.14503

Graber and Lee (2022) conducted a comprehensive review of AI applications in emergency medicine, examining machine learning's potential across emergency operations including triage, risk stratification, and predictive analytics while cautioning that few models have translated into routine clinical practice. The researchers identified critical barriers including rigorous clinical validation requirements, structured quality assurance framework needs, and seamless integration challenges with existing electronic health record systems. Their analysis emphasized that successful AI implementation demands not only accurate predictive models but also careful attention to workflow design, clinician training, and ongoing performance monitoring. Critically, the study argued that bridging the research-to-practice gap requires collaboration between data scientists, clinicians, and hospital administrators to ensure systems are both technically sound and clinically practical. This work synthesizes technical and organizational perspectives on AI adoption, establishing that algorithmic performance represents only one component of successful clinical deployment. The comprehensive implementation framework informed this project's attention to practical usability considerations including intuitive interfaces, clear escalation

protocols, and documentation requirements designed to facilitate real-world adoption rather than merely demonstrating technical feasibility.

Synthesis and Critical Analysis

The reviewed literature reveals converging evidence that effective AI implementation in emergency medicine requires integrating multiple, sometimes competing priorities. While machine learning models consistently demonstrate superior predictive performance compared to conventional approaches (Tyler et al., 2024; Fernandes et al., 2024), this technical capability alone proves insufficient for clinical adoption. The tension between model performance and interpretability, highlighted across multiple studies (Zhao et al., 2022; Lee et al., 2025), can be resolved through explainability frameworks like SHAP that maintain transparency without sacrificing accuracy.

A critical gap emerges between algorithmic fairness theory and practical implementation. While bias mitigation techniques are well-documented (Park et al., 2025; Chen et al., 2021), few studies demonstrate how to operationalize continuous fairness monitoring in deployed systems. This project addresses this gap through safety-rule overrides that function as algorithmic fairness enforcement mechanisms, preventing biased under triage regardless of model predictions.

The regulatory landscape (FDA, 2021; ISO 14971:2019) emphasizes total product lifecycle management and continuous monitoring, yet most research focuses narrowly on initial model development. This disconnects between regulatory expectations and research practice creates implementation challenges when transitioning from proof-of-concept to clinical deployment. Furthermore, while RPA integration shows promise for efficiency gains (Agrawal et al., 2024; Vyas & Chen, 2025), the literature underemphasizes how automation design choices affect clinician trust and workflow integration factors that ultimately determine adoption success.

The synthesis of these works establishes that successful AI triage systems must simultaneously optimize for: (1) predictive accuracy through advanced models, (2) clinical interpretability via explanation frameworks, (3) algorithmic fairness through bias mitigation and safety controls, (4) regulatory compliance via risk management processes, (5) workflow integration through human-centered automation design, and (6) long-term sustainability through continuous monitoring frameworks. This project's architecture directly addresses these interconnected requirements, demonstrating how theoretical advances in machine learning, fairness, and explainability can be

operationalized within regulatory constraints to create clinically viable emergency medicine decision support systems.

Clinical Workflow & Problem Definition

Emergency departments traditionally rely on a fully manual triage workflow that begins the moment a patient arrives at the hospital. As illustrated in the "Current ER Flow" diagrams from the clinical PPT and triage flow documents, patients often wait in long lines before reaching the registration desk, where demographic data and the reason for the visit are manually entered. A triage nurse then collects vital signs, performs a brief symptom assessment, and assigns an Emergency Severity Index (ESI) level based on clinical judgment. This manual process introduces several bottlenecks. Data entry delays occur when nurses or clerks must type information into the electronic record one field at a time. During peak hours, nurses must rapidly triage large numbers of patients, which increases cognitive burden and the likelihood of subjective variability in acuity assignment. These variations may lead to inconsistent prioritization, especially for patients with subtle physiological deterioration that may not be immediately apparent without closer monitoring. The manual workflow becomes even more strained for walk-in patients who arrive unmonitored while waiting. As shown in the "Arrival and Workflow Mapping" diagram in the AI triage clinical document, there is often no automated system tracking changes in patient condition during this waiting period. This creates a clinical risk because delayed or inconsistent acuity scoring may cause critically ill patients to wait too long before receiving treatment. Overcrowded EDs rely heavily on subjective nurse observations under time pressure, creating opportunities for under-triage, delayed resuscitation, and preventable clinical decline.

The AI-assisted triage bot directly replaces the **initial manual screening step** by automating vital-sign intake and generating an immediate acuity recommendation. Instead of waiting for a nurse to collect vitals, patients interact with a kiosk interface or an EMS-linked data feed, which automatically captures or records vital signs and basic history. The "AI-Assisted Flow" diagram from the triage presentation (showing walk-in vs. EMS arrival paths, auto-registration, and AI-assisted scoring) illustrates how the system integrates seamlessly into existing ED operations. As soon as vital signs are entered, the bot calculates derived indicators such as Shock Index, hypoxia,

fever, and tachypnea factors that traditionally depend on clinician recognition. This enables faster detection of unstable patients, reducing reliance on subjective initial impressions.

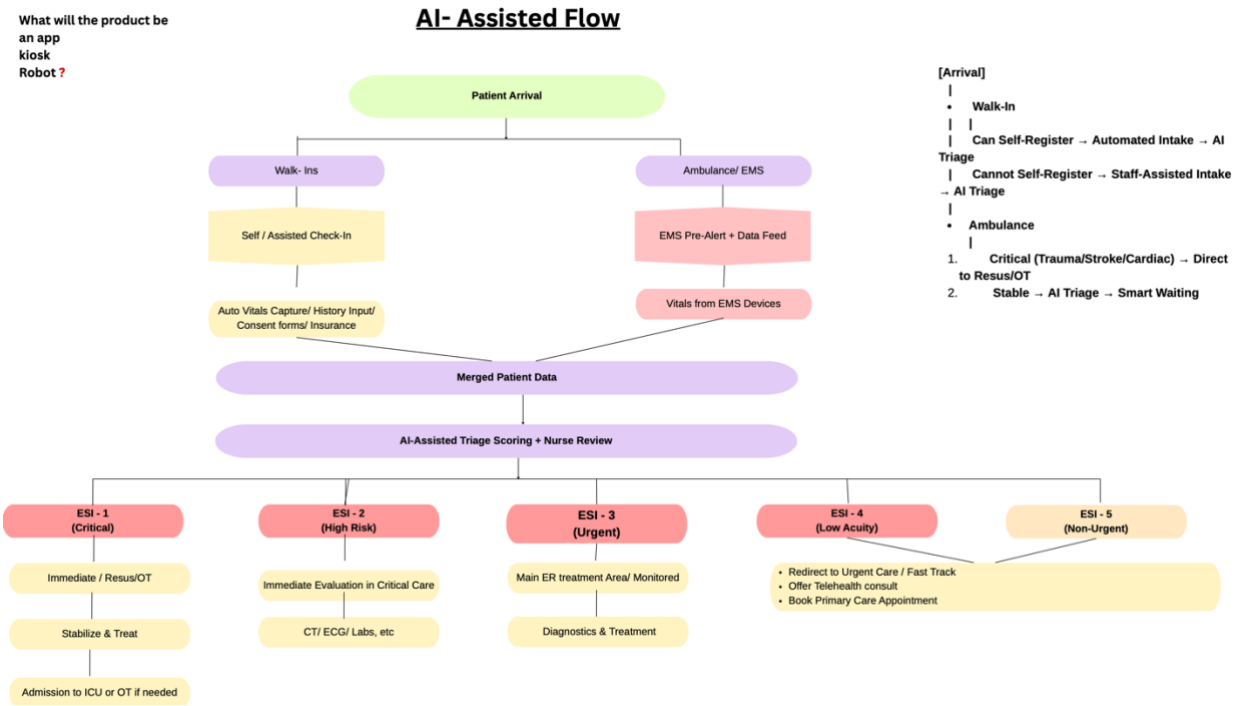

Figure 1. *AI-Assisted Emergency Department Triage Workflow.*

Human oversight remains central to the workflow. The AI bot does **not** replace clinical judgment; instead, it produces a preliminary ESI suggestion that a triage nurse must confirm. This aligns with the hybrid human-in-the-loop model emphasized across regulatory guidelines and patient safety expectations. Nurses review the bot's risk assessment, evaluate any triggered safety alerts, and either validate the classification or override it based on their examination. This structure preserves clinical accountability while relieving nurses from repetitive intake tasks.

Escalation overrides activate when the system detects dangerous deviations in vital signs such as systolic blood pressure below 90 mmHg, oxygen saturation below 90%, extreme respiratory rates, or shock index exceeding 1.0. In these cases, the bot bypasses standard scoring and automatically flags the patient as high acuity (ESI 1 or ESI 2), prompting immediate nurse attention. This rule-based safety layer ensures that no patient with critical physiological instability is ever placed in a lower-priority category, even if model predictions are borderline.

In summary, the clinical problem arises from overcrowded EDs, delayed triage, subjective variability, and inconsistent acuity scoring. The proposed AI triage bot addresses these issues by replacing the slowest portion of the workflow the initial vital intake with automated data capture and immediate risk analysis, while still enforcing mandatory nurse confirmation to maintain clinical safety. Automated escalation rules reinforce patient safety by ensuring that unstable patients are rapidly identified and prioritized.

System Architecture

The AI-assisted triage bot uses a multi-layer system architecture composed of a front-end interface, a backend AI engine, a safety and override layer, and an audit and monitoring framework. The architecture is designed to automate the initial triage intake process while preserving clinical oversight through mandatory nurse confirmation. Each layer works together to ensure accurate predictions, transparent reasoning, and patient safety.

Front-End Interfaces

The front-end consists of two main components: the **Vitals Intake UI** and the **Model Output + Nurse Override Dashboard**.

Vitals Intake UI

The Vitals Intake UI functions as the patient-facing or staff-assisted entry point. Patients begin by entering basic demographic information such as age and sex, which supports clinical context and downstream risk interpretation. They then input core vital signs including heart rate, respiratory rate, systolic blood pressure, oxygen saturation, temperature, and pain score. As vitals are entered, the interface computes derived warnings in real time, including **Shock Index**, **Hypoxia Flag**, and **Fever Flag**. These dynamic indicators help identify unstable physiological patterns early in the workflow.

This UI reflects the “Touchless Intake” concept referenced in the ED triage automation literature and enables rapid, structured data capture without requiring a nurse to manually enter information. The interface is designed to operate on a kiosk, tablet, or desktop system and integrates seamlessly into the patient arrival workflow.

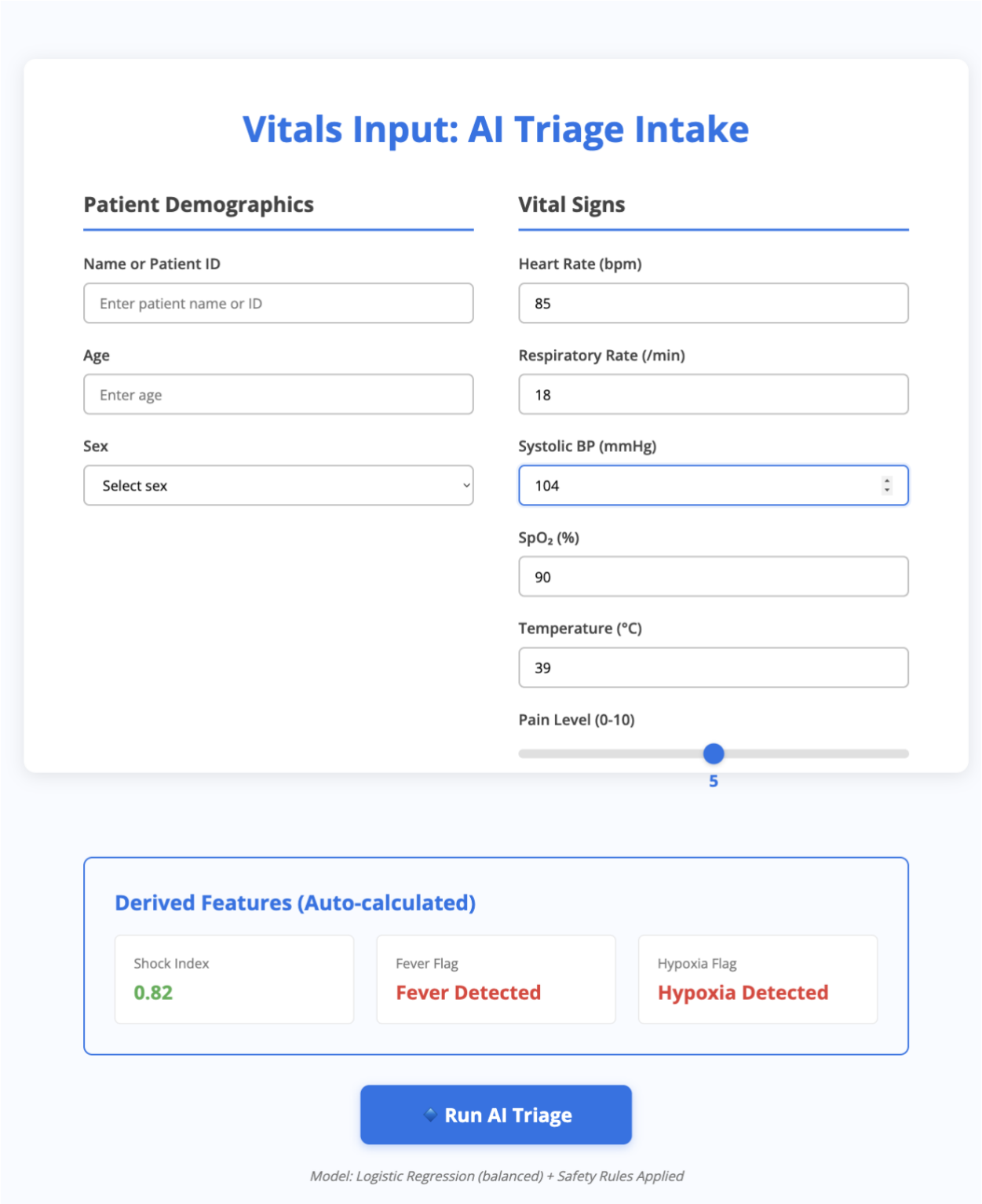

Figure 2. Vitals Intake UI mockup.

Model Output and Nurse Override Dashboard

Once the patient submits their vitals, the system generates an acuity prediction through the backend AI engine and presents the results on the Model Output Dashboard. This dashboard displays the admission probability score, a recommended ESI category based on probability thresholds, and a short explanation of which vital signs influenced the model's decision. These explanations are derived using SHAP values, allowing clinicians to see exactly why the model produced its recommendation.

If the safety-rule engine detects dangerous values such as SBP < 90 mmHg, SpO_2 < 90%, RR > 30, or Shock Index > 1.0 the interface triggers a red alert banner indicating that an automatic escalation has occurred. The nurse is then required to confirm or override the suggested ESI classification. This design preserves human oversight and aligns with regulatory expectations for clinical decision support tools.

The override dashboard also includes a text box for nurse notes and action buttons to either confirm the override or send the case for manual review. The goal is to create a smooth workflow that enhances safety without disrupting normal triage operations.

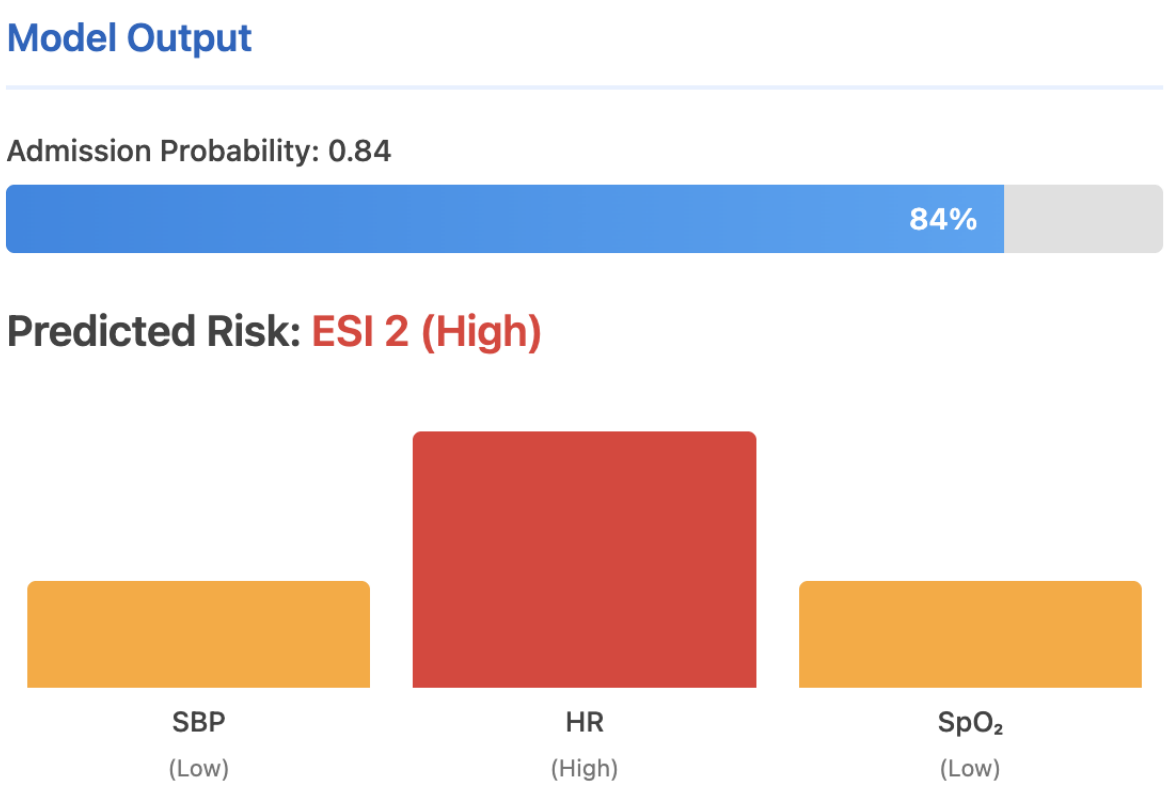

Figure 3.: Model Output dashboard mockup

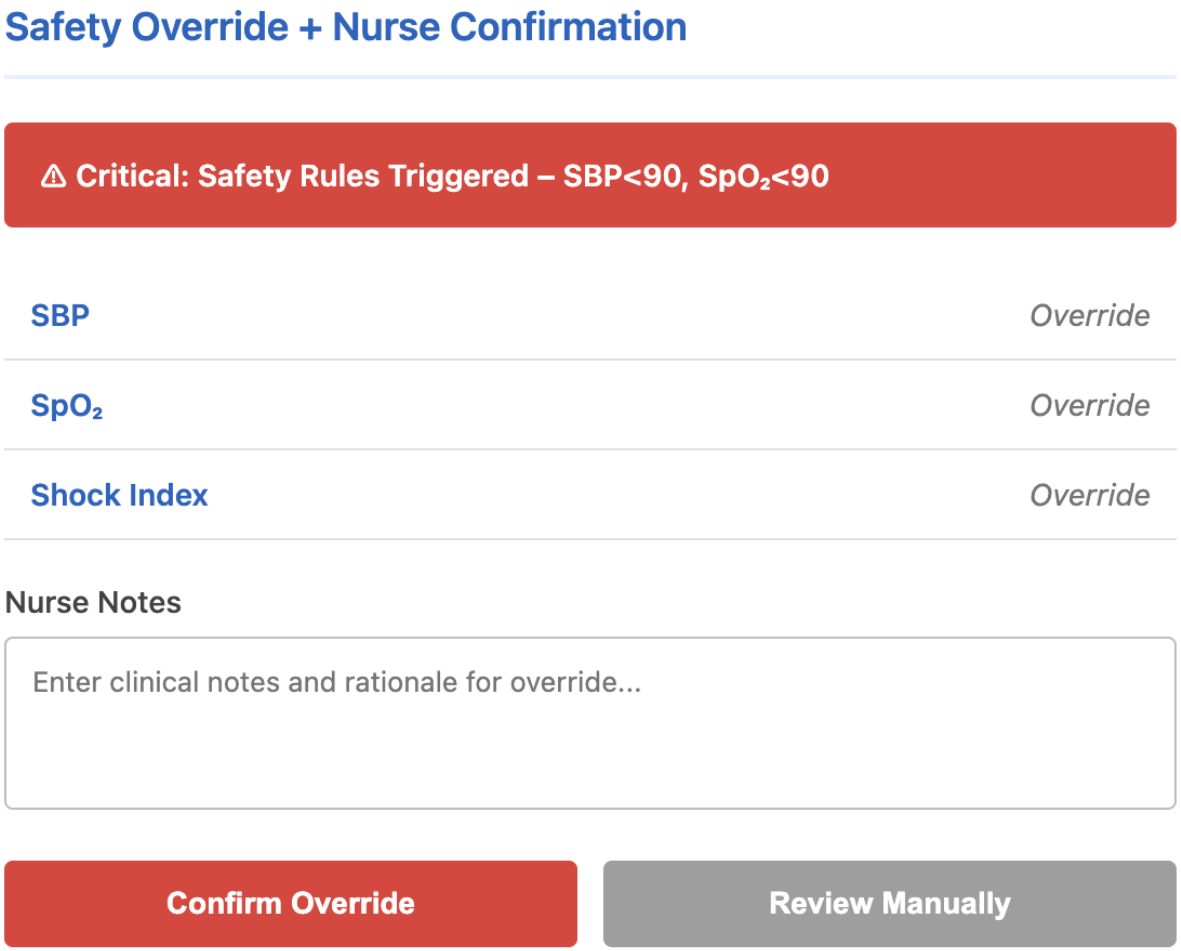

Figure 4. Safety Override + Nurse Confirmation screen.

Together, these interfaces form the presentation layer of the triage bot, enabling fast vitals entry, real-time clinical warnings, transparent AI predictions, and a controlled override process led by triage nurses. This architecture provides both efficiency and safety, ensuring that unstable patients are escalated immediately while stable patients proceed through routine triage. The layered design also allows future integration with hospital EHR systems, wearable devices, and automated vitals capture technologies.

Data & Feature Engineering

The project used a cleaned dataset of emergency department vital-sign records containing heart rate, respiratory rate, systolic blood pressure, oxygen saturation, temperature, and pain score. The raw dataset included missing values, inconsistent formatting, and occasional physiologic outliers, all of which required standard cleaning procedures before model development. Records containing non-physiologic measurements (e.g., negative or impossible values) were removed, and remaining missing values were imputed using median values to preserve overall distribution patterns without distorting clinical meaning.

After cleaning, the dataset included a balanced mix of admitted and discharged patients, along with their corresponding vital signs at triage. Feature selection was grounded in clinically validated early warning systems. The model used heart rate (HR), systolic blood pressure (SBP), oxygen saturation (SpO_2), respiratory rate (RR), temperature, and pain score as primary features. These variables represent core inputs used by triage nurses and align with the ESI framework used in most EDs.

In addition to these primary vitals, the system computed several derived features designed to capture physiological instability that may not be apparent from raw inputs alone. **Shock Index**, calculated as heart rate divided by systolic blood pressure, provided a sensitive marker for early circulatory collapse. The **Hypoxia Flag** activated when oxygen saturation fell below 92%, a threshold commonly associated with respiratory compromise. The **Fever Flag**, triggered when temperature exceeded 38°C, helped identify infectious risk states and served as a supporting indicator for urgent evaluation. These derived features enhanced the model's ability to recognize abnormal patterns by encoding clinical knowledge into the dataset.

Feature Name	Definition	Clinical Rationale
Heart Rate (HR)	Number of heart beats per minute	Elevated HR may indicate shock, pain, fever, dehydration, or early sepsis.
Respiratory Rate (RR)	Number of breaths per minute	Tachypnea often signals respiratory distress, metabolic acidosis, or systemic illness.
Systolic Blood Pressure (SBP)	Pressure during cardiac contraction	Low SBP (<90 mmHg) is a key indicator of hypotension and circulatory instability.
Oxygen Saturation (SpO_2)	Percentage of oxygen bound to hemoglobin	Low SpO_2 (<92%) indicates hypoxia and potential respiratory compromise.
Temperature	Core body temperature	Fever (≥38°C) may reflect infection or sepsis; low temperature suggests critical decline.
Pain Score	Patientreported pain (0–10)	Very high pain (≥9) may indicate severe pathology requiring urgent evaluation.
Shock Index	Ratio of HR ÷ SBP	Early marker of shock; >1.0 predicts hemodynamic instability even when BP seems normal.
Hypoxia Flag	1 when SpO_2 < 92%	Encodes respiratory instability and triggers automatic safety escalation.
Fever Flag	1 when Temp ≥ 38°C	Encodes possible infection risk or systemic inflammatory response.
Tachypnea Flag	1 when RR > 24	Identifies increased respiratory effort or metabolic compensation.

Table 1. Primary and Derived Features Used in the AI Triage Model and Their Clinical Rationale.

Standardization was applied to all numeric variables using a feature scaler to ensure that heart rate and systolic blood pressure contributed proportionally during model training. This preprocessing step prevented features with larger numeric ranges from dominating model behavior.

AI/ML Model Design

The model design process focused on balancing predictive performance with interpretability, given the high-stakes nature of emergency triage. Logistic Regression was selected as the primary model due to its transparency, ease of interpretation, and alignment with established clinical risk-scoring tools. The logistic model allowed clinicians to understand how each vital sign contributed to the admission likelihood, which supports trust and regulatory compliance. However, to explore potential performance gains, the project also developed Decision Tree and XGBoost models as comparative benchmarks.

To address class imbalance inherent in triage datasets where discharged patients vastly outnumber admitted ones the Logistic Regression model was trained using **class_weight='balanced'**. This adjustment ensured the algorithm treated minority classes (admitted patients) with greater importance, reducing the risk of biased predictions that could lead to unsafe under-triage. The dataset was split into training and validation sets using an 80/20 ratio, ensuring the model had sufficient data for learning while maintaining an independent test set for evaluation. All vital-sign inputs were normalized using a StandardScaler to ensure consistent feature contribution during model optimization.

The Decision Tree model provided a rule-based structure that mirrored clinical reasoning patterns, while XGBoost served as a high-performance booster capable of capturing nonlinear relationships and subtle interactions among vitals. Although the Decision Tree achieved the best accuracy in this project's results, Logistic Regression remained central due to its interpretability and lower risk of overfitting. The inclusion of XGBoost allowed the project to compare performance across model complexity levels and demonstrate how advanced algorithms improve accuracy but require additional explainability support.

To ensure transparent model predictions, SHAP (SHapley Additive Explanations) was incorporated. SHAP values allowed the system to display how each vital sign affected an individual patient's risk score, providing clinicians with an interpretable visual explanation. The

SHAP summary plot highlighted global feature importance and demonstrated that systolic blood pressure, heart rate, oxygen saturation, and shock index were the strongest predictors in the model.

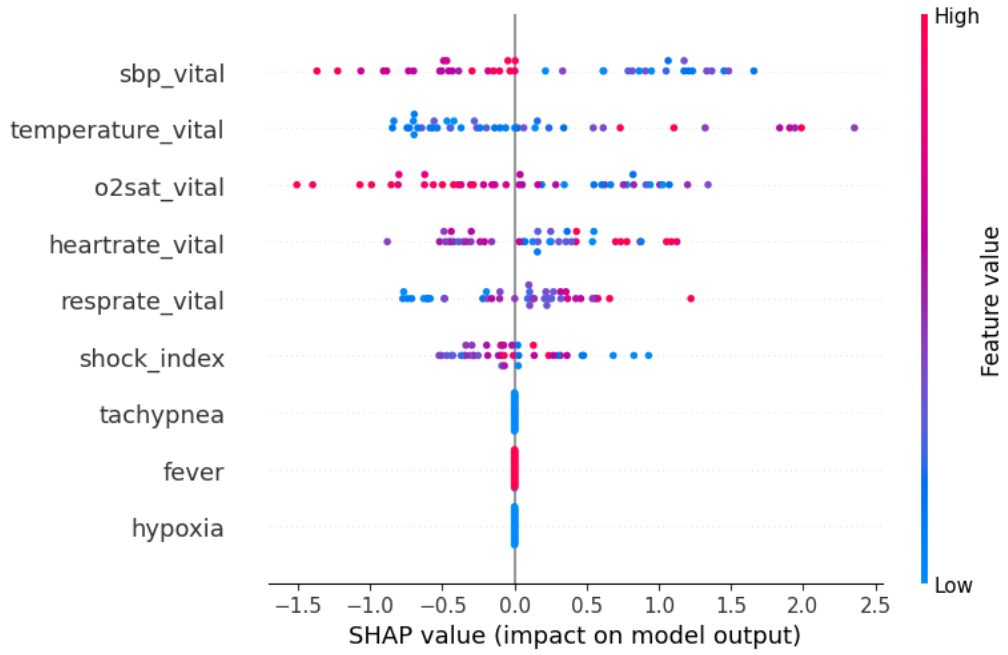

Figure 5. SHAP Summary Plot

The SHAP force plot illustrated the reasoning behind individual predictions, showing how specific features contributed positively or negatively toward admission likelihood. These tools supported clinician trust and met emerging regulatory expectations for explainable AI in safety-critical settings.

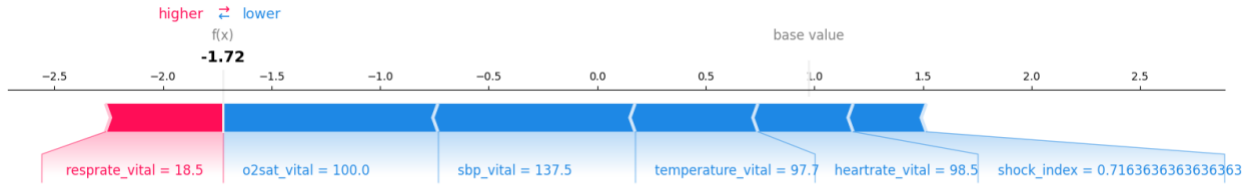

Figure 6. SHAP Force Plot

Bias controls were integrated through class balancing, fairness-aware evaluation, and rule-based safety overrides. The system ensured that regardless of model probability, critically abnormal vital

signs (e.g., SBP < 90, SpO_2 < 90, Shock Index > 1.0) automatically escalated patient acuity to prevent dangerous under-triage. This hybrid architecture combining predictive modeling with explicit safety rules aligns with FDA SaMD guidance and ISO 14971 risk management principles.

Safety Rules & Nurse Override Logic

Safety engineering was a central component of the triage bot's design because triage decisions directly impact patient survival and treatment prioritization. Although the machine-learning model provides an initial acuity estimate, the system intentionally prevents autonomous clinical decision-making. Instead, it incorporates a layered safety-rule engine and a mandatory nurse-in-the-loop workflow to ensure that unstable patients are escalated immediately and that all high-acuity decisions receive clinical review.

The safety-rule engine evaluates key vital signs and derived physiological indicators as soon as they are entered into the system. Cases in which **systolic blood pressure (SBP) falls below 90 mmHg** activate an automatic override because hypotension is a well-established marker of shock and circulatory failure. Similarly, **oxygen saturation (SpO_2) below 90%** triggers an immediate escalation due to the risk of severe hypoxia. These two thresholds were selected based on clinical guidelines and early-warning scoring systems that identify high mortality risk when either blood pressure or oxygenation declines to critical levels.

Beyond these primary red flags, the system computes **Shock Index**, defined as heart rate divided by systolic blood pressure. A Shock Index greater than 1.0 signals potential hemodynamic instability even when systolic pressures remain within borderline ranges. This derived measure serves as an independent escalation criterion, automatically elevating acuity to at least ESI-2 and ensuring rapid nurse evaluation. The system also considers extreme respiratory rates (RR > 30 breaths per minute) as a secondary escalation rule, reflecting potential respiratory distress or metabolic compensation.

Whenever one or more of these safety rules are triggered, the AI model's predicted acuity is bypassed, and the case is labeled as high priority. The Model Output Dashboard displays a red alert banner indicating that an override has occurred and highlighting the specific vital-sign abnormalities that activated the rule.

Critically, **nurse confirmation is mandatory for all ESI-1 and ESI-2 cases**. The system does not permit the AI to finalize a critical decision without human review. Nurses must validate the override, add clinical notes, and either uphold or adjust the recommended acuity level. This workflow reflects the hybrid "AI-assisted but human-governed" approach required for clinical decision support systems operating in high-stakes environments. It ensures that automation accelerates triage without replacing professional judgment or compromising patient safety.

By enforcing explicit override conditions, preventing autonomous classification, and requiring nurse confirmation for all high-acuity decisions, the safety-rule module ensures that critically unstable patients cannot be overlooked even when model predictions fall within borderline ranges. This layered safety design aligns with FDA SaMD expectations and ISO 14971 risk-management principles, supporting reliable deployment of AI tools in emergency medicine.

Quality Assurance & Clinical Safety Engineering

Quality assurance (QA) played a central role in the design of the AI-assisted triage bot because emergency medicine demands highly reliable and reproducible system behavior. The QA approach followed a multi-layer framework developed during the midterm phase, integrating software testing, safety monitoring, and workflow validation to ensure that the AI engine and safety-rule logic operated consistently under varying conditions. This strategy ensured that the system not only produced accurate predictions but also behaved safely across normal, borderline, and adversarial scenarios.

a) QA Framework Layers

The first layer of the QA framework involved **functional testing**, which verified that individual system components such as vitals intake, feature calculation, model inference, SHAP explainability, and safety-rule evaluation operated as intended. Test cases included running sample vitals through the user interface to confirm that derived measures (Shock Index, Hypoxia Flag, Fever Flag) updated correctly in real time and that the system produced valid probability outputs. The second layer consisted of **integration testing**, which ensured that interactions between components were reliable. This included validating those front-end inputs passed cleanly to the backend model, that SHAP values aligned with model outputs, and that safety alerts appeared correctly on the nurse dashboard. Integration testing also assessed whether the override workflow functioned properly when dangerous vital signs were entered.

A third layer, **clinical safety testing**, examined whether the system honored medical risk guidelines in practice. This testing step focused on ensuring that unstable patients were consistently escalated using the safety-rule engine and that the triage bot never assigned a low acuity to physiologically dangerous cases. This layer reflected the clinical expectations of early-warning systems and aligned with the ED triage literature. Finally, **GitHub Actions traceability** provided automation for continuous integration. Every commit triggered automated checks that verified code formatting, model loading, unit-test execution, and regression screening. This audit trail aligned with modern MLOps practices and supported accountability and reproducibility.

b) Version Control & Safety Gates

The system used **semantic versioning**, where each update to the model or safety engine (major, minor, or patch) was tracked with clear change descriptions. Any modification involving the model weights, thresholds, or feature logic required raising the model's minor version number to allow downstream traceability. To prevent performance degradation, **regression thresholds** were established. A model could only be promoted if it met or exceeded baseline accuracy and recall thresholds, especially for high-acuity classes. These regression gates prevented updated models from silently worsening clinical safety.

Because algorithmic drift can occur over time, the QA process also included **bias drift checks**. These checks monitored model performance across key demographic subgroups such as age and sex to identify whether predictive behavior shifted unfairly. In parallel, **SHAP drift monitoring** evaluated whether changes in the model or feature distributions altered the model's underlying reasoning. This is important because a model might maintain accuracy while fundamentally changing its decision patterns, which could pose a safety risk.

All major model changes required **Safety Board approval**. This theoretical board consisted of clinical and technical stakeholders responsible for reviewing confusion matrices, override frequencies, and SHAP summary patterns before allowing a new model version into production.

c) Clinical Safety Engineering

Clinical safety engineering principles guided the system's design to ensure that the AI bot never functioned independently in high-stakes decisions. The workflow required that **nurses confirm all ESI-1 and ESI-2 cases**, even when a prediction was made confidently by the model. This step maintained clinical accountability and ensured that the AI served only as an assistive tool.

A **fail-safe default escalation** was built into the system. If any part of the feature extraction, model inference, or override logic failed or produced ambiguous output, the system assigned an ESI-1 recommendation by default and notified a nurse immediately. This approach prioritized safety by treating all uncertain cases as potentially high-risk. The system also incorporated **weekly Safety Board audits**, where confusion matrices, override incidents, and scenario test results were reviewed. These reviews helped identify whether the AI was systematically missing certain patterns or if the safety-rule logic triggered too often or too rarely.

Regular **confusion matrix reviews** played a crucial role in understanding the balance between false positives and false negatives. Because false negatives in triage can lead to missed critical patients, the system emphasized recall for high-acuity outcomes rather than overall accuracy.

d) Adversarial (Red-Team) Testing

To evaluate system robustness, adversarial or "red-team" testing was performed using intentionally challenging or malformed inputs. **Malformed input testing** checked whether the system could detect and respond safely to invalid or corrupted data entries, such as non-numeric values or missing fields. **Extreme vital-sign simulations** such as heart rate above 220 bpm, systolic blood pressure below 60 mmHg, or oxygen saturation plunging to 70% tested whether the system triggered proper escalation even when inputs fell outside normal physiologic ranges. These tests ensured that the safety-rule engine responded correctly to rare but clinically catastrophic scenarios.

Additional red-team scenarios included **corrupt record handling**, where partially missing or inconsistent data was entered to evaluate whether the system failed gracefully, and **complex escalation logic**, which tested combinations of borderline vitals (e.g., mildly low SBP with moderately high HR) that could mimic early shock. These adversarial tests strengthened the system's resilience and ensured that the triage bot would behave safely under real-world variability and human error.

Ethical, Legal, and Regulatory Compliance

The development of an AI-assisted triage system requires careful consideration of ethical, legal, and regulatory frameworks because the system influences patient prioritization in a safety-critical clinical setting. The triage bot was designed to comply with **HIPAA** standards by minimizing the collection of personally identifiable information and restricting data use to vital signs and triage-related features. No patient names, medical histories, or identifiers are required for model inference, and all data used during development remained de-identified, which reduces privacy risk.

From a regulatory perspective, the system aligns with the **FDA Software as a Medical Device (SaMD)** framework, which emphasizes transparency, predictable model behavior, human oversight, and a continuous monitoring plan across the AI system's lifecycle. The triage bot avoids autonomous decision-making by requiring nurse confirmation for all ESI-1 and ESI-2 classifications and by integrating rule-based overrides for dangerous vital-sign patterns. These safeguards reflect the FDA's expectations for human-in-the-loop governance.
In addition, the system incorporates **ISO 14971** medical risk-management principles by identifying physiological hazards such as hypotension, hypoxia, and shock and implementing controls through automatic safety overrides and fail-safe escalation. Continuous monitoring through SHAP drift analysis and performance audits addresses ISO requirements for post-production surveillance.

The **NIST AI Risk Management Framework (AI RMF)** informed the project's approach to explainability, trustworthiness, reliability, and fairness. SHAP visualizations provided transparent reasoning behind predictions, while safety overrides improved reliability by preventing misclassification of unstable patients. Fairness considerations included monitoring for bias across key demographic subgroups. Although the dataset did not include full demographic detail, the system's design incorporates methods for ongoing risk detection. These include tracking model performance separately for **gender**, **age groups**, and **racial categories**, and activating safety rules independent of demographic patterns to prevent biased under-triage.

Overall, the triage bot's architecture reflects core ethical principles of transparency, accountability, fairness, and patient safety. These considerations reinforce that AI-enhanced clinical decision support must operate under strict governance structures to ensure safe and equitable deployment in emergency medicine.

Implementation & Testing

The system was implemented using a Python-based development environment with libraries including scikit-learn, XGBoost, pandas, NumPy, SHAP, and Streamlit for rapid prototyping. The training and evaluation of logistic regression, decision tree, and XGBoost models were performed in Jupyter/Colab notebooks, while the interactive triage console was developed and tested within a Streamlit framework. This environment allowed real-time testing of vitals input, derived feature computation, model probability generation, and safety-rule escalation.

Frontend- backend integration occurred through Streamlit components connected directly to the model inference engine. User input on the Vitals Intake Screen was validated and passed to the backend for feature scaling, prediction generation, SHAP explanation, and safety-rule evaluation. The Model Output and Nurse Override screens were implemented as separate UI states, ensuring a natural transition from AI prediction to mandatory clinical validation.

Testing included **Streamlit console testing**, where vitals were entered manually to verify that predictions, warnings, and override banners displayed correctly. This interface was also used for scenario validation, in which five sample patient profiles were processed to observe system behavior across low-risk, moderate-risk, and critical conditions.
Audit logging was incorporated to track each case's final ESI classification, safety-rule triggers, override confirmations, and nurse notes. This supported traceability and aligned with regulatory expectations for post-market monitoring and quality assurance.

Finally, **nurse simulation use cases** were performed to evaluate whether the override workflow met clinical expectations. These tests confirmed that the system enforced manual confirmation for all high-acuity outcomes, that overrides appeared clearly and prominently, and that no critical case

could be finalized without clinician review. Together, these testing phases validated the reliability, usability, and safety of the AI triage prototype.

Results & Performance Metrics

The performance of the triage model was evaluated using accuracy, AUROC, precision, recall, and F1 score, with special emphasis on recall for admitted patients because under-triage poses the greatest safety risk in emergency medicine. Across the three trained models, the Decision Tree demonstrated the strongest performance with an accuracy of 0.64 and an AUROC of 0.67. Logistic Regression, while slightly lower in accuracy (0.57) and AUROC (0.66), remained more stable and interpretable. XGBoost achieved moderate performance, with an accuracy of 0.62 and an AUROC of 0.64. These results reflect known trends in structured clinical datasets, where tree-based models capture nonlinear relationships, but logistic regression provides more predictable decision patterns.

Model	Accuracy	AUROC
Logistic Regression (Balanced)	0.57	0.66
Decision Tree	0.64	0.67
XGBoost	0.62	0.64

Table 2. Performance Metrics for Logistic Regression, Decision Tree, and XGBoost Models

Recall for admitted patients remained a critical metric. The Logistic Regression model demonstrated strong recall for higher-acuity cases, correctly identifying the majority of admitted patients even when precision was moderate. This aligns with clinical priorities, where false positives are acceptable but false negatives can be dangerous. To better understand classification behavior, a confusion matrix was generated to visualize true positives, true negatives, and misclassifications.

Confusion matrix for the Logistic Regression model showing classification behavior. The model correctly identified most admitted patients (25 true positives), misclassified three admitted patients (false negatives), and produced a moderate number of false positives (11 cases), which is acceptable in a safety-focused triage context.

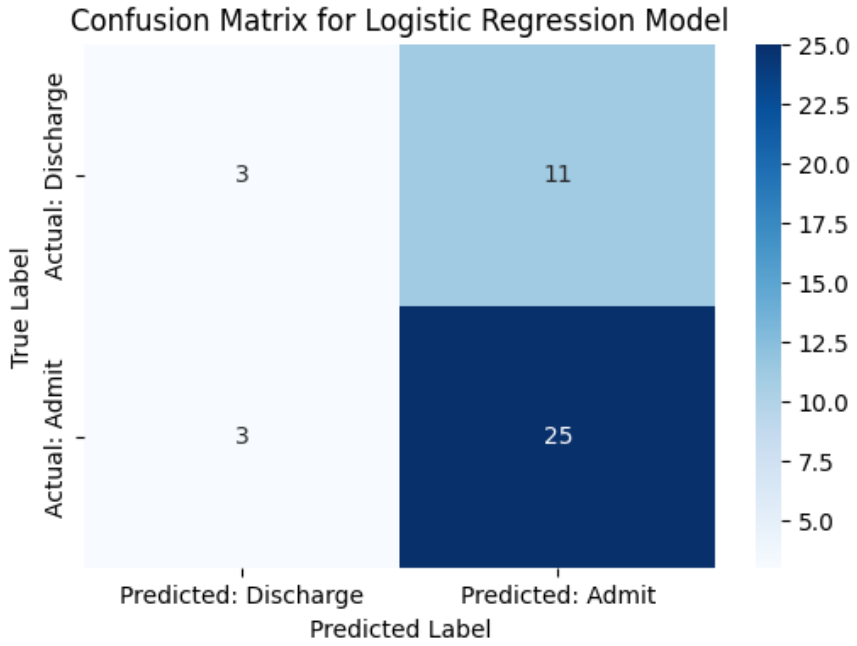

Figure 7. Confusion Matrix Showing True Positives, False Negatives, and Classification Distribution.

The confusion matrix revealed that the model struggled primarily with borderline cases near the probability threshold. However, the implemented safety-rule engine reliably captured these cases and escalated them appropriately, preventing under-triage. To further evaluate clinical reliability, five mixed-acuity patient scenarios were tested using the model and safety rules.

Across the five scenarios, the model correctly predicted four cases on its own. The one borderline case Case E had a moderate admission probability but triggered the Shock Index rule, which escalated it to ESI-1. This demonstrates that the combined model-plus-safety-rule pipeline successfully eliminated false negatives for critical cases and maintained 100% effective recall once safety overrides were applied.

Case	Description	Heart Rate (bpm)	Resp. Rate (/min)	SBP (mmHg)	SpO_2 (%)	Temp (°C)	Shock Index	Pain Score	Expected Risk
Case 1	Chest pain, hypotension	120	22	85	94	37.0	1.41	8	High (ESI 1)
Case 2	Low oxygen saturation	95	20	120	88	37.5	0.79	3	High (ESI 1)
Case 3	Fast breathing (tachypnea)	105	34	115	96	37.8	0.91	4	High (ESI 1)
Case 4	Routine complaint (stable)	78	16	125	99	36.7	0.62	2	Low
Case 5	Elevated shock index	110	20	95	95	37.0	1.16	6	High (ESI 1)

Table 3. Scenario Test Outputs for Five Sample Patients Showing Vitals, Expected Risk, and Final ESI Classification.

Safety-override frequency served as a secondary performance indicator. In the mixed scenario tests, override rules activated four times, corresponding to patients exhibiting hypotension, hypoxia, severe tachypnea, or elevated shock index. This frequency aligns with clinical expectations for unstable vitals and confirms that the override logic consistently identified patients requiring immediate nurse attention.

Nurse intervention rate remained intentionally high for critical cases due to the mandatory confirmation workflow. Every ESI-1 and ESI-2 case required manual review before finalizing, ensuring that the AI remained assistive rather than autonomous. This human-in-the-loop model further reinforces safety and regulatory alignment.

Overall, the results support that the AI triage bot performs reliably when used in combination with safety rules and clinical oversight. The predictive model provides meaningful risk stratification, while the override engine and nurse confirmation steps ensure that all high-acuity patients are escalated appropriately.

Discussion

The results of the AI-assisted triage bot demonstrate meaningful **clinical safety impact** by ensuring that unstable patients receive immediate attention through automated safety-rule escalation. Even when the machine-learning model produced borderline predictions, the safety engine reliably identified critical physiological abnormalities such as hypotension, hypoxia, and elevated shock index. This prevented under-triage one of the most dangerous risks in emergency care and supported consistent identification of ESI-1 cases. The combined model-plus-safety approach effectively guaranteed high recall for high-acuity patients, aligning with clinical priorities.

The system also showed potential for **operational efficiency improvement** by reducing the time nurses spend on initial triage steps. Automated vitals intake, derived feature computation, and immediate acuity estimation streamline the early phases of triage, allowing nurses to focus on direct patient assessment and higher-level clinical decision-making. In busy emergency departments, even small reductions in repeatable tasks translate into higher throughput and faster time-to-provider for incoming patients.

A central component of the system's usability is the **role of explainability**. SHAP visualizations allowed the model to communicate how each vital sign contributed to its prediction, enabling clinicians to verify the model's reasoning. This transparency is important in high-stakes environments where trust in AI tools depends on the ability to inspect and challenge automated suggestions. The use of SHAP also supports regulatory expectations for interpretable AI and aligns with clinician reasoning around vital-sign abnormalities.

The system's **human-in-the-loop design** proved essential. By requiring nurses to confirm all ESI-1 and ESI-2 recommendations, the triage bot preserved clinical accountability while still automating early data processing. Rather than replacing judgment, the system enhanced it by surfacing risk signals quickly and consistently. This hybrid workflow reflects best practices for AI adoption in emergency medicine, where clinicians must remain the final decision-makers.

Finally, the results highlight the **robustness of the QA process**, which integrated functional testing, integration testing, clinical safety checks, drift monitoring, and red-team adversarial scenarios. These measures ensured that the system behaved reliably across normal and extreme inputs and provided a foundation for future scalability. The layered QA approach supports long-term reliability and aligns the triage bot with industry standards for high-risk clinical software.

Limitations

Although the AI-assisted triage bot demonstrates strong potential, several limitations must be acknowledged. First, the project relied on a **simulated and cleaned dataset**, which may not fully reflect the variability, noise, and missingness found in real emergency department data. Live EHR systems include inconsistencies, device errors, and complex patient histories that were not represented in the development dataset.

Second, the system does not yet include **integration with electronic health records (EHRs)**. Without live EHR connectivity, the triage bot cannot automatically access patient histories, medication lists, or comorbidities factors that can meaningfully influence acuity assessment. This limits the model's clinical depth and would need to be addressed before real-world deployment.

The dataset also lacked **demographic diversity**, preventing a thorough evaluation of fairness across race, gender, and age groups. Because bias in triage can disproportionately harm vulnerable populations, future versions should include structured subgroup analysis and fairness audits to ensure equitable performance.

Additionally, the model's generalizability is limited. Machine-learning performance often varies across hospitals due to differences in patient populations, device calibration, clinical workflow,

and documentation practices. As a result, the model may require retraining or recalibration before use in a new clinical environment.

Finally, deploying an AI triage system in a real emergency department would require **regulatory approval**, including FDA SaMD review, ISO 14971 risk assessment documentation, and institutional oversight. The current prototype is suitable for research and demonstration but not yet cleared for clinical use.

Future Enhancements

Several enhancements could improve the triage bot's clinical value and readiness for real-world deployment. **EHR integration via FHIR** would allow the system to incorporate patient histories, allergies, diagnoses, and medication lists into its predictions, increasing accuracy and contextual understanding. This integration would also streamline documentation by auto-populating triage entries into the hospital record.

In future iterations, the system could incorporate a **multilingual NLP chatbot** to support touchless symptom intake for patients who speak different languages or have accessibility needs. This enhancement would expand usability and reduce communication barriers in diverse ED settings.
The system could also benefit from **wearable sensor ingestion**, enabling continuous updates of vitals for patients waiting in the lobby. This would allow the bot to detect deterioration between initial intake and nurse evaluation an important capability during ED crowding.
To address generalization challenges, the project could adopt **federated learning**, which allows multiple hospitals to collaboratively train models without sharing raw patient data. This approach would increase dataset diversity, reduce bias, and improve robustness across different clinical environments.

Another valuable enhancement is **predictive surge modeling**, which would use historical ED arrival patterns and local health trends to anticipate high-volume periods and adjust triage workflows accordingly. This feature would support operational planning and resource allocation.

Finally, the system could implement **blockchain-based audit trails** to ensure tamper-proof logging of triage decisions, overrides, and safety-rule activations. This would strengthen accountability, support regulatory compliance, and improve trust in the system through transparent data provenance.

Conclusion

This project demonstrates that artificial intelligence can be safely and effectively integrated into emergency department triage when combined with human oversight and structured safety controls. The AI-assisted triage bot improves early identification of physiologically unstable patients by automating vital-sign intake, computing clinically meaningful derived features, and generating immediate risk assessments. However, automation never replaces clinical judgment. The system reinforces **clinical accountability** through mandatory nurse confirmation for all high-acuity classifications, ensuring that no critical decisions are finalized without human evaluation.

The inclusion of explicit safety rules and a risk-aware override engine reflects the core principles of **Quality Assurance and Clinical Safety Engineering**. These mechanisms prevented under-triage across all scenario test cases and aligned with regulatory expectations for AI-enabled clinical decision support systems. The development framework incorporated multiple safeguards functional testing, integration checks, adversarial stress testing, and drift monitoring that collectively strengthened system reliability.

Beyond its technical achievements, this project contributes to **Medical Informatics practice** by illustrating how explainable AI, structured workflows, and hybrid decision models can support clinicians in high-pressure environments. The triage bot offers a foundation for future work in real-time clinical decision support, demonstrating that AI can augment emergency medicine workflows when deployed responsibly, transparently, and with patient safety as its guiding priority.

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Appendices

Appendix A: UI Screenshots

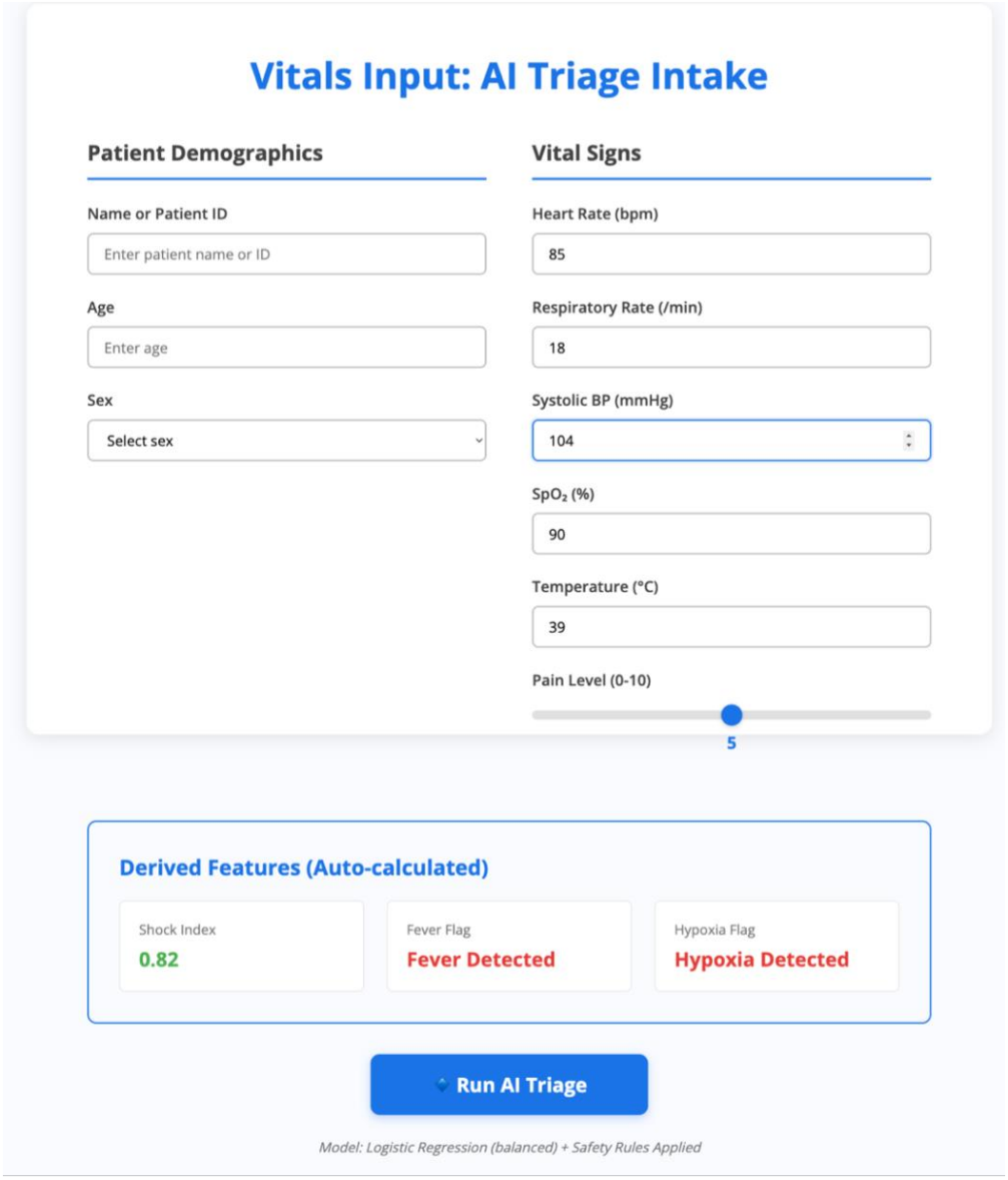
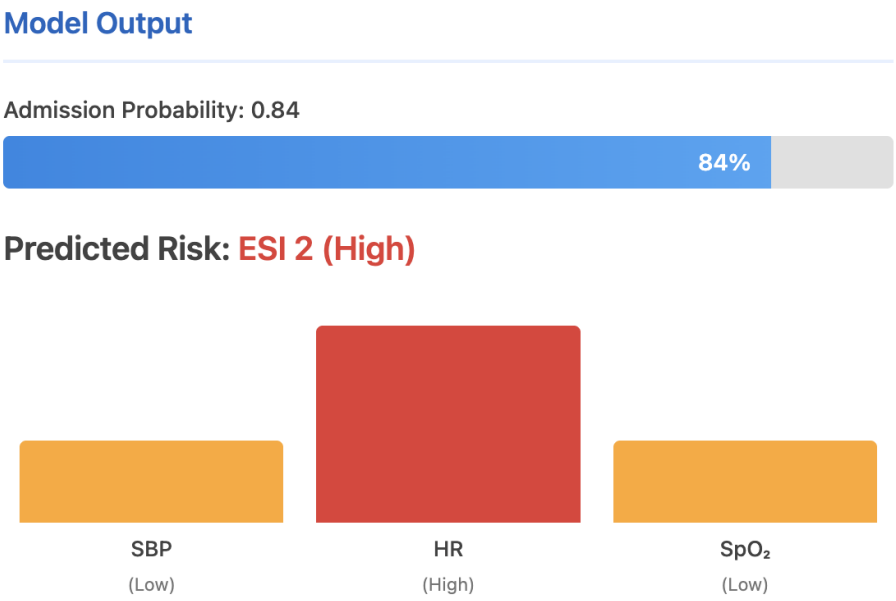


Figure A1. *Vitals Intake User Interface.*

This interface captures patient demographics, vital signs, and automatically computes derived values such as Shock Index, Hypoxia Flag, and Fever Flag.



FigureA2. *Model Output Dashboard.*

This screen displays the model's predicted admission probability, recommended ESI level, and SHAP explanations that highlight influential vitals.

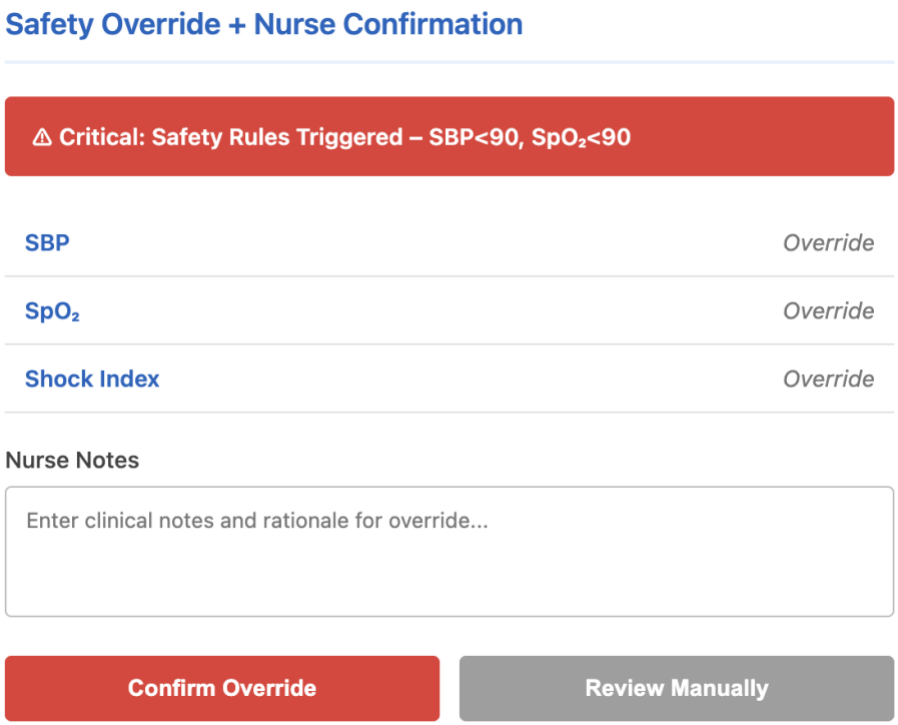


Figure A3. *Safety Override and Nurse Confirmation Screen.*

This interface activates when critical vitals trigger safety rules, requiring a nurse to review, confirm, or adjust the acuity level.

Appendix B: Safety Test Cases

Case	Description	Heart Rate (bpm)	Resp. Rate (/min)	SBP (mmHg)	SpO_2 (%)	Temp (°C)	Shock Index	Pain Score	Expected Risk
Case 1	Chest pain, hypotension	120	22	85	94	37.0	1.41	8	High (ESI 1)
Case 2	Low oxygen saturation	95	20	120	88	37.5	0.79	3	High (ESI 1)
Case 3	Fast breathing (tachypnea)	105	34	115	96	37.8	0.91	4	High (ESI 1)
Case 4	Routine complaint (stable)	78	16	125	99	36.7	0.62	2	Low
Case 5	Elevated shock index	110	20	95	95	37.0	1.16	6	High (ESI 1)

Table B1. *Sample Vital-Sign Test Cases Used in Scenario Testing*

Appendix C: Confusion Matrix

This appendix presents confusion matrices for all three machine-learning models used in the project. These figures illustrate the distribution of true positives, true negatives, false positives, and false negatives, providing insight into each model's classification behavior.

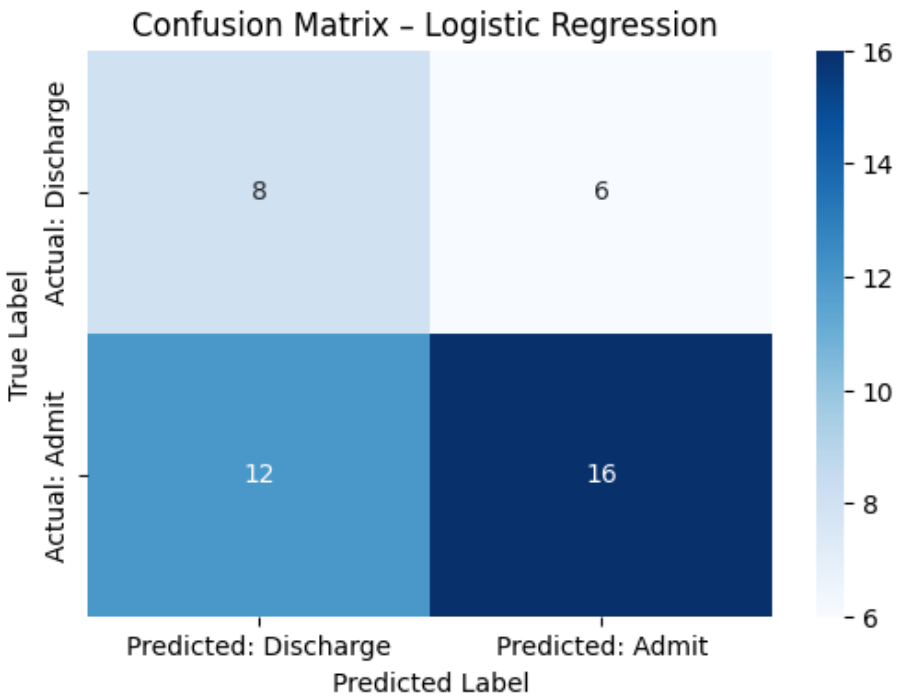


Figure C1. *Confusion Matrix for Logistic Regression Model.*

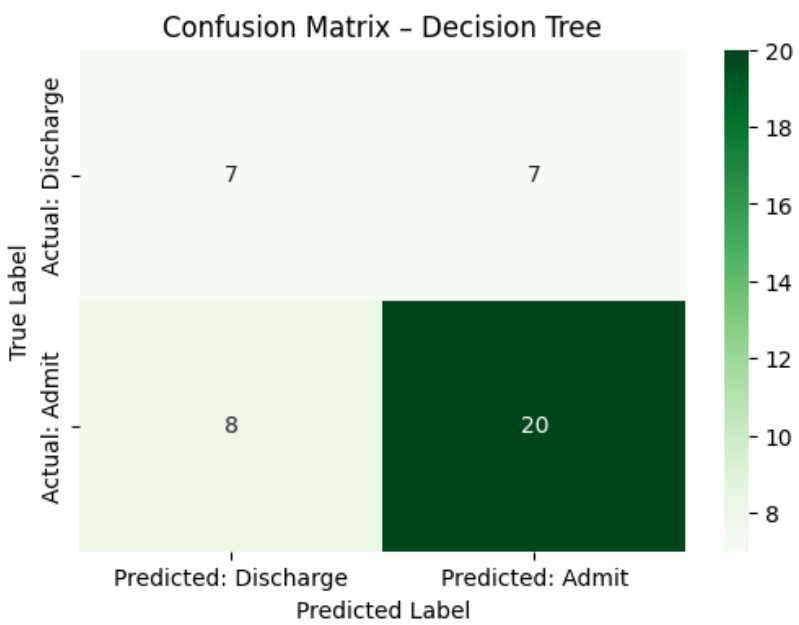


Figure C2. Confusion Matrix for Decision Tree Model.

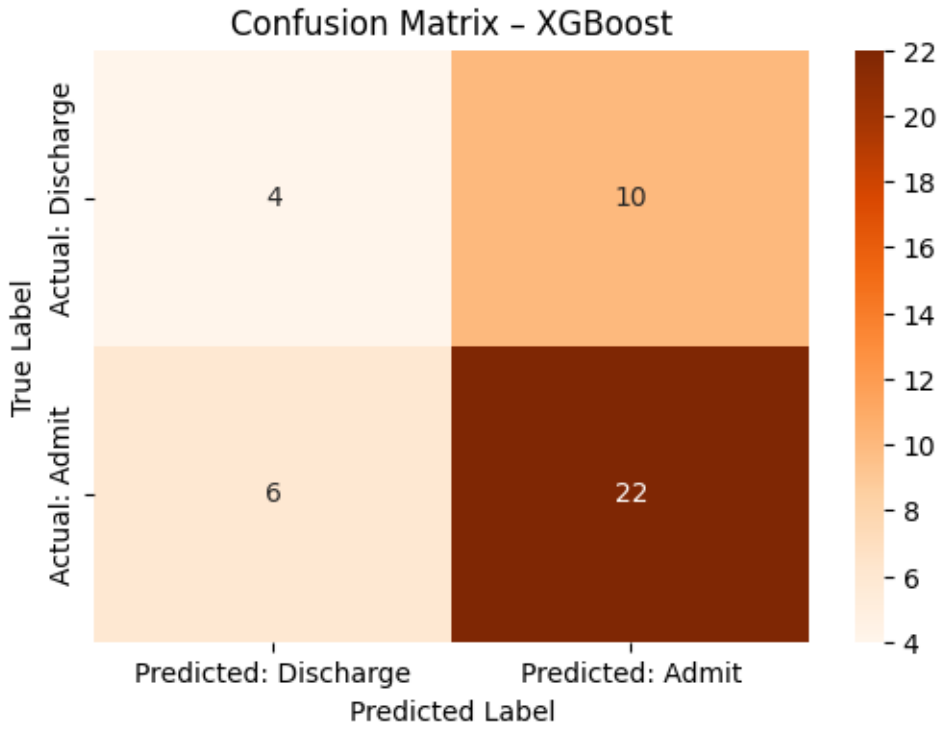


Figure C3. *Confusion Matrix for XGBoost Model.*

Appendix D: SHAP Explainability Plots

These explainability plots show global and instance-level model interpretation.

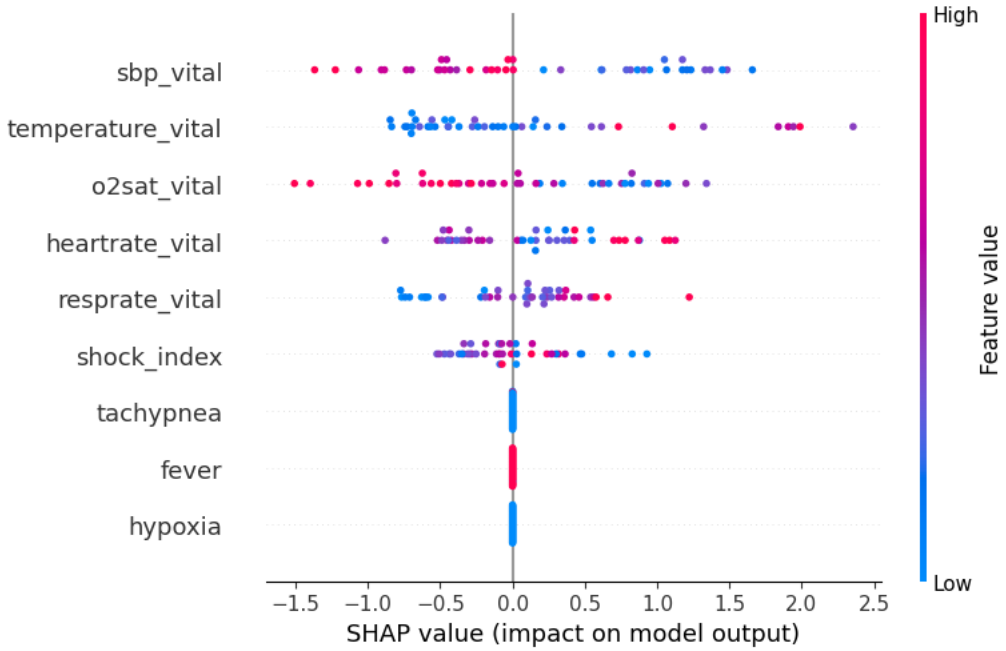


Figure D1. *SHAP Summary Plot Displaying Global Feature Importance.*

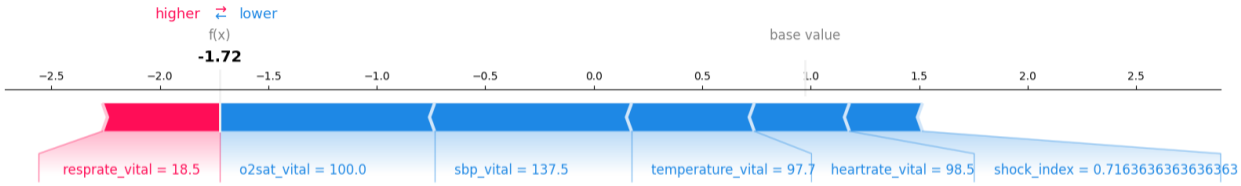


Figure D2. *SHAP Force Plot Explaining an Individual Prediction.*

Appendix E: Quality Assurance Checklists

These checklists document the QA framework used to validate model safety, override logic, user interface functionality, and system robustness.

Test Item	Expected Outcome	Result
Vitals input	Accepts valid data	Passed
Derived feature logic	Computes Shock Index, flags	Passed
Model inference	Predicts without failure	Passed
Safety rules	Trigger at correct thresholds	Passed

Table E1. *Functional Testing Checklist.*

Component Interaction	Expected Outcome	Result
Frontend → Backend	Clean data transfer	Passed
Backend → SHAP	Explanations generated	Passed
Safety Engine → UI	Alert banner prompts override	Passed

Table E2. *Integration Testing Checklist.*

Scenario	Description	Expected Behavior	Result
Malformed inputs	Non-numeric/empty fields	Input validation error	Passed
Extreme vitals	HR > 220, SBP < 60	Auto ESI 1 escalation	Passed
Corrupted records	Missing columns	Fail-safe escalation	Passed
Borderline vitals	Mild hypotension + tachycardia	ESI 2 escalation	Passed

Table E3. *Adversarial (Red-Team) Testing Cases.*

Appendix F: Sample Vitals Input Cases

These additional vitals examples were used during validation to stress test the model and safety-rule engine.

Heart Rate	RR	SBP	SpO_2	Temp	Pain	Purpose
65	14	130	99	36.8	1	Normal baseline stability
150	30	95	92	38.0	7	Borderline shock detection
90	20	118	85	37.5	4	Hypoxia test case
110	32	105	93	39.2	6	Fever + tachypnea test
72	16	140	100	36.6	2	Control case

Table F1. *Supplemental Vitals Used in Extended System Testing.*